

Homework Assignment 2

1. Monte Carlo Integration. Suppose the volume V_d of the d -dimensional unit ball is known. In this problem, use a d -dimensional Monte Carlo integral to estimate the volume V_{d+1} of the unit ball in $(d+1)$ dimensions. By slicing the $(d+1)$ -dimensional ball at $\mathbf{x} = (x_1, \dots, x_d)$,

$$V_{d+1} = 2 \int_{|\mathbf{x}| \leq 1} \sqrt{1 - |\mathbf{x}|^2} d^d x.$$

This provides a simple benchmark for Monte Carlo integration, since the exact ball volumes are known.

You may use either of the following sampling approaches:

Method A: Sample uniformly inside the d -dimensional unit ball B_d . Using the sampling method discussed in lecture,

$$V_{d+1} = 2V_d \left\langle \sqrt{1 - |\mathbf{x}|^2} \right\rangle_{B_d} \approx 2V_d \frac{1}{N} \sum_{\substack{i=1 \\ \mathbf{x}_i \sim \text{Unif}(B_d)}}^N \sqrt{1 - |\mathbf{x}_i|^2}.$$

Method B: Sample uniformly inside the cube $[-1, 1]^d$. Define

$$g(\mathbf{x}) = \begin{cases} \sqrt{1 - |\mathbf{x}|^2}, & |\mathbf{x}| \leq 1, \\ 0, & |\mathbf{x}| > 1. \end{cases}$$

Then

$$V_{d+1} = 2 \times 2^d \langle g(\mathbf{x}) \rangle_{[-1,1]^d} \approx 2 \times 2^d \frac{1}{N} \sum_{\substack{i=1 \\ \mathbf{x}_i \sim \text{Unif}([-1,1]^d)}}^N g(\mathbf{x}_i).$$

Method B is less efficient because points outside the unit ball make zero contribution.

Tasks

- Pick one dimension $d \geq 2$ and write a Monte Carlo code to evaluate the integral above using either Method A or Method B. Higher-dimensional calculations will be considered for extra credit.
- The first goal of this exercise is simply to obtain a good numerical estimate of the integral. By the end of today's in-class coding session (or today's submission deadline), submit your **best estimate** of V_{d+1} , together with enough numerical output to show that your code is working.
- Next, look more carefully at how the error decreases as the number of Monte Carlo samples N increases. Make a log-log plot of the error versus N and check whether the slope is approximately $-1/2$. The exact unit-ball volumes are

d	2	3	4	5	6	7	8
V_d	π	$\frac{4\pi}{3}$	$\frac{\pi^2}{2}$	$\frac{8\pi^2}{15}$	$\frac{\pi^3}{6}$	$\frac{16\pi^3}{105}$	$\frac{\pi^4}{24}$

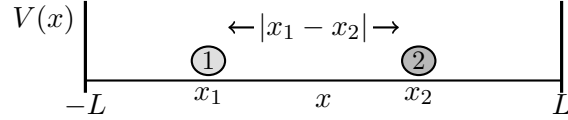
2. Rejection Sampling: Two Interacting Particles. Consider two identical charged particles constrained to move in a one-dimensional box. Let their positions be x_1 and x_2 . The total potential energy is

$$E(x_1, x_2) = V(x_1) + V(x_2) + U(x_1 - x_2),$$

where

$$V(x) = \begin{cases} 0, & -L < x < L, \\ \infty, & \text{otherwise,} \end{cases} \quad U(y) = \frac{e^2}{|y|}.$$

Here the Coulomb prefactor is absorbed into e^2 .



At temperature T , the equilibrium configurational probability density is

$$\pi(x_1, x_2) = \frac{1}{Z} e^{-\beta E(x_1, x_2)}, \quad \beta = \frac{1}{k_B T}.$$

Inside the box,

$$\pi(x_1, x_2) \propto \exp \left[-\frac{\beta e^2}{|x_1 - x_2|} \right].$$

Introduce the dimensionless interaction strength $\lambda = \beta e^2 / L$. In this problem take $L = 1$.

Tasks

- For $\lambda = 1$, use rejection sampling to generate equilibrium configurations (x_1, x_2) . Make a **2D histogram (probability-density heatmap)** in the (x_1, x_2) plane and briefly describe its main features. Also report the acceptance rate $N_{\text{accepted}}/N_{\text{proposed}}$.
- For two noninteracting particles independently and uniformly distributed in $[-L, L]$,

$$\langle |x_1 - x_2| \rangle_0 = \frac{2L}{3}.$$

Numerically compute $\langle |x_1 - x_2| \rangle$ for at least $\lambda = 0.5, 1.0, 1.5, 2.0$ (you are encouraged to include additional values of λ if you wish), and plot $\langle |x_1 - x_2| \rangle$ as a function of λ . Include the noninteracting value ($\lambda = 0$) as a reference point and briefly discuss the trend.